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Review

X-in-the-Loop Methodology for Proton Exchange Membrane Fuel Cell Systems Design: Review of Advances and Challenges

Hugo Lambert [1], David Hernàndez-Torres [1], Clément Retière [2], Laurent Garnier [1] and Jean-Philippe Poirot-Crouvezier [1,*]

[1] Univ. Grenoble Alpes, CEA, LITEN, 38000 Grenoble, France; hugo.lambert@cea.fr (H.L.); david.hernandeztorres@cea.fr (D.H.-T.); laurent.garnier@cea.fr (L.G.)
[2] CEA, CEA Tech Pays de la Loire, F-44340 Bouguenais, France; clement.retiere@cea.fr
* Correspondence: jean-philippe.poirot@cea.fr

Abstract

Proton Exchange Membrane Fuel Cells (PEMFCs) are seen as an alternative for heavy-duty transportation electrification. Powered by a green hydrogen source, they can provide high efficiency and low carbon emissions compared to traditional fuels. However, to be competitive, these systems require high reliability when operated in real-life conditions, as well as safe and efficient operating management. In order to achieve these goals, the X-in-the-loop (also called model-based design) methodology is well suited. It has been largely adopted for PEMFC system development and optimisation, as they are complex multi-component systems. In this paper, a systematic analysis of the scientific literature is conducted to review the methodology implementation for the design and improvement of the PEMFC systems. It exposes a precise definition of each development step in the methodology. The analysis shows that it can be employed in different ways, depending on the subsystems considered and the objectives sought. Finally, gaps in the literature and technical challenges for fuel cell systems that should be addressed are identified.

Keywords: PEMFC; HIL; hardware-in-the-loop; real-time testing; transportation

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1. Introduction

In order to comply with the Paris COP21 agreement, the electrification of the transport sector is now occurring in major markets (Europe, North America, and Asia). Light transport is converting from internal combustion engines to hybrid and full electrical vehicles. Fuel cell vehicles can offer long range, fast recharge times, and no pollution during driving. However, they suffer from low efficiency compared to battery electrical vehicles. Indeed, the efficiency of a fuel cell electrical vehicle ranges between 50 % and 65 % [1,2] (from the energy delivered by the hydrogen tank to the wheels), while the efficiency of a battery electric vehicle can be close to 90 % [3]. Nevertheless, in heavy-duty transportation, such as maritime applications, public transport, or goods trucks, hydrogen (H_2) could be an interesting energy carrier. For a high-frequency response and energy recovery, fuel cells are hybridised with high-power storage devices, such as batteries and supercapacitors. In this case, the fuel cell serves as the main energy source, while the high-power storage device acts as a transient power supply and can recover energy during braking phases, thus improving overall efficiency.

When integrated into a vehicle, the fuel cell is surrounded by auxiliary devices, which together form a fuel cell system (FCS). The FCS balance of plant is responsible for electrical

power conditioning through appropriate power converters. It also ensures the proper conditioning and supply of hydrogen and air in terms of pressure, humidity, and flow, and guarantees suitable thermal management to avoid the overheating of the fuel cell, which could lead to degradation. When these systems operate in real-life conditions, they must work efficiently, safely, durably, and reliably to be competitive.

Among the various areas where improvements must be made to develop FCS integration into real-world applications, system durability remains crucial. In 2021, a durability of 17,000 h under 20 % degradation was recorded by the U.S. Department of Energy on a real-world fuel cell bus [4]. The ultimate target is set at 25,000 h, representing a 50 % improvement in durability. In Europe, reliability issues have occurred in some fuel cell trains, leading local authorities to suspend operations on the world's first H_2 train line in Germany [5]. These are the two main concerns for large-scale integration. Other topics of major concern for proper FCS integration in real-world applications include the following:

- FCS efficiency;
- The system's ability to operate in a degraded mode;
- Optimal energy and power management of the FCS;
- FCS health prognosis and management.

All of these issues can be addressed by the model-based design approach, or the so-called X-in-the-loop (XIL) methodology, where X represents various terms such as "hardware", "software", or "model", which will be listed and explained in detail in this paper.

This framework is inherited from the automotive sector, where vehicle complexity is constantly rising with increasing parts of electronics, software, algorithms, and autonomous systems. Traditionally, the safety and the reliability of the system was determined by the components' fidelity. With the rise of automation, software reliability must be tested. The development of real-time testing tools enables the use of the XIL methodology and a transition from the previous classical approach (modelling and then direct testing on the real system) to some intermediary steps. These includes software-in-the-loop validation and rapid control prototyping, that enable software development testing directly on the system under examination. And finally, there is hardware-in-the-loop testing that consists of testing together the real system (classically the controller) with virtual sub-systems in real-time. This is part of a strategy to reduce risks by testing only some parts of a system [6]. This approach, widely used in the automotive sector, can be applied for the development and optimisation of FCSs, as it reduces risks and the time for the development of new technologies. This paper presents a systematic review of the scientific literature to investigate how the XIL methodology can help enhance FCSs and facilitate their integration into the transport sector. It provides an overview of the studies published in the academic literature. It is worth noting that this review does not refer to patents, as they generally provide limited experimental proof and are not peer-reviewed. Moreover, the availability of commercial products demonstrates that complete development processes have been applied to fuel cell systems by industries. Eventually, they have applied XIL procedures, which have not been disclosed.

In the following section, the XIL methodology is detailed, focusing on the specificity of this approach in its application to academic research compared to industry. The role of each procedure within the general approach is described, and a detailed definition is systematically proposed. The second section discusses the contributions of this methodology to the development and integration of fuel cell systems, classifying the published works regarding the different subsystems of the FCS they are studying. Subsequently, the challenges that require attention are highlighted and explored in the third section. This research concludes with a summary of the key findings and insights.

2. XIL Methodology and Definitions

In this section we present the XIL methodology workflow as well as the dedicated terminology. To help the reader's understanding, some references are proposed as example applications of the XIL framework. More specific PEMFC applications are discussed in Section 3.

2.1. Historical Review and Terminology

The hardware-in-the-loop (HIL) approach was initially used in the aerospace industry as a flight simulator [7] and has since been widely applied in the automotive industry. However, the increasing complexity of controllers in the automotive industry has driven the sector to develop the XIL methodology [8]—where X stands for model (M), software (S), or hardware (H)—enabling the transition from modelling to real-time validation. This methodology is also known as "model-based design" (MBD) with the V-cycle development process [9], as shown in Figure 1.

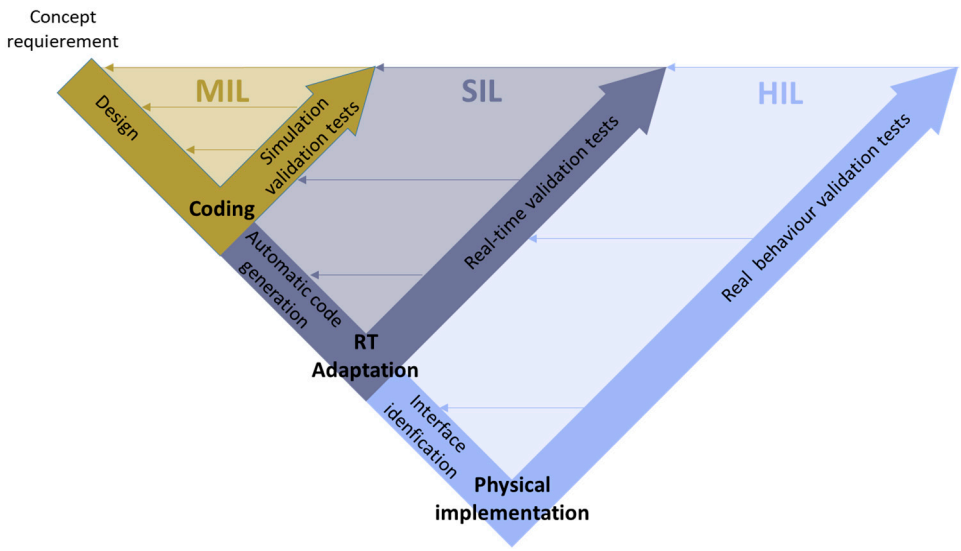

Figure 1. Classical industrial V-cycle used for the description of the XIL or MBD workflow.

The process begins with model-in-the-loop (MIL) for design and simulation validation tests. Next is the software-in-the-loop (SIL) step, where the code is compiled, and real-time constraints are addressed using techniques that will be presented in the next section. Finally, HIL testing involves interface identification and the physical implementation of real-world hardware with real-time simulation. At each step of the development process, it is possible to make adjustments to previous steps. This workflow facilitates the implementation and validation of control systems and reduces development time, cost, and risks.

In the same context, we also find Rapid Control Prototyping (RCP), which is widely used for testing new algorithms on real physical systems. RCP testing can be used for validating control algorithms on prototypes, for instance in the case of a code update, in a fast and efficient way, without the necessity of code exportation into embedded controller. This is possible thanks to specifics tools that are real-time computers with a lot of input and outputs possibilities that allows us to stay in the development environment. We can cite manufacturers like Speedgoat, OPAL-RT, and dSPACE. RCP shares similar characteristics with HIL. However, as part of a V-cycle workflow, the HIL approach includes successive validations that are not present in the RCP approach. The objective of RCP is to quickly demonstrate a proof-of-concept for control among other applications.

Researchers from various academic fields have attempted to adopt these methods for their work. However, the wide range of applications and varying requirements has led to

different terminologies and their associated meanings within the academic community, potentially causing misunderstandings. This section proposes a clarification of the semantics for XIL methodologies. We begin by distinguishing between simulation and emulation. These two concepts differ in their level of abstraction of the system. Simulation relies on pure numerical models to mimic the behaviour of a system, ranging from a single equation to complex digital twins. In contrast, emulation uses real hardware to replicate the flows or signals of the real system and its environment. Table 1 compares the academic and the automotive points of view on the XIL methodology. The most common abbreviations are MIL (Model-In-the-Loop), SIL (Software-In-the-Loop), HIL (Hardware-In-the-Loop), P-HIL (Power-Hardware-In-the-Loop), C-HIL (Controller-Hardware-In-the-Loop), and RCP (Rapid-Control-Prototyping). They are defined in detail in the next chapter.

Table 1. Definition of the XIL acronym in automotive industry and in academic world.

XIL	Automotive World Point of View	Proposed Academic Point of View	Difference
MIL	Development of control laws in a simulated environment, more or less faithfully, without real-time constraints	Simultaneous code development for the control laws and for the model of the system without real-time constraints	The automotive world uses pre-existing models of the system
SIL	Real-time PC-based testing of control functions using an environment simulator (vehicle)	Adaptation of the code of the control laws and of the model of the system to meet real-time constraints: model reduction, fixed time steps, code compilation	The automotive world uses a PC platform with pre-existing compiled models of the system called simulators
HIL	Testing of the actual microcontroller interfaced with a real-time simulated environment	All tests involving a hardware or emulated element interfaced with a real-time simulated system	The automotive world restricts this approach to microcontroller validation
P-HIL	Testing of the actual microcontroller interfaced with physical parts and real-time emulated parts of the system under test (SUT) with power exchanges	Emulation of one or possibly multiple physical devices controlled by a real-time model through a test-bench involving power exchanges. It can be connected to a real physical system (actuator, engine, or battery). Only a part of the system may be emulated, while the other is simulated. This is referred to as semi-virtual simulation	The automotive world restricts this approach to microcontroller validation
C-HIL	-	Testing of the actual microcontroller interfaced with a real-time simulated environment	The academic "C-HIL" corresponds to the automotive "HIL"
RCP	Development of control laws interfacing with a real physical system	Development of control laws interfacing with a real physical system	Similar definitions

Other terms occasionally appear in the literature in this field. FIL (FPGA-In-the-Loop) [10] designates specific cases of SIL where the controller operates on an FPGA (Field Programmable Gate Array) for very fast commands [11], or on a processor. FIL also allows the use of FPGAs to emulate systems with very fast behaviours (less than 10 µs), such as power converters [12]. PIL (Processor-In-the-Loop) [13] corresponds to deploying the controller code or the real-time system model on a dedicated processor [14]. In academic terminology, it is one of the steps within SIL. CIL (Controller-In-the-Loop) [15] covers testing activities similar to C-HIL [16]. Finally, the term RIL (Real-Physics-In-the-Loop) [17] is sometimes used to refer to the general use of physical or electrical actuators. In our academic terminology, it can be grouped under the more commonly used term P-HIL. Other terms designating XIL methods are used more marginally and are not detailed here. For the rest of this paper, only the academic point of view is considered, as the references are from the academic field.

Finally, the different stages of XIL can be represented based on their physical support, with the exception of RCP, which is defined by its specific objective. A graphical representation is provided in Figure 2. This figure denotes the interactions between the virtual and real worlds in each step of the XIL process.

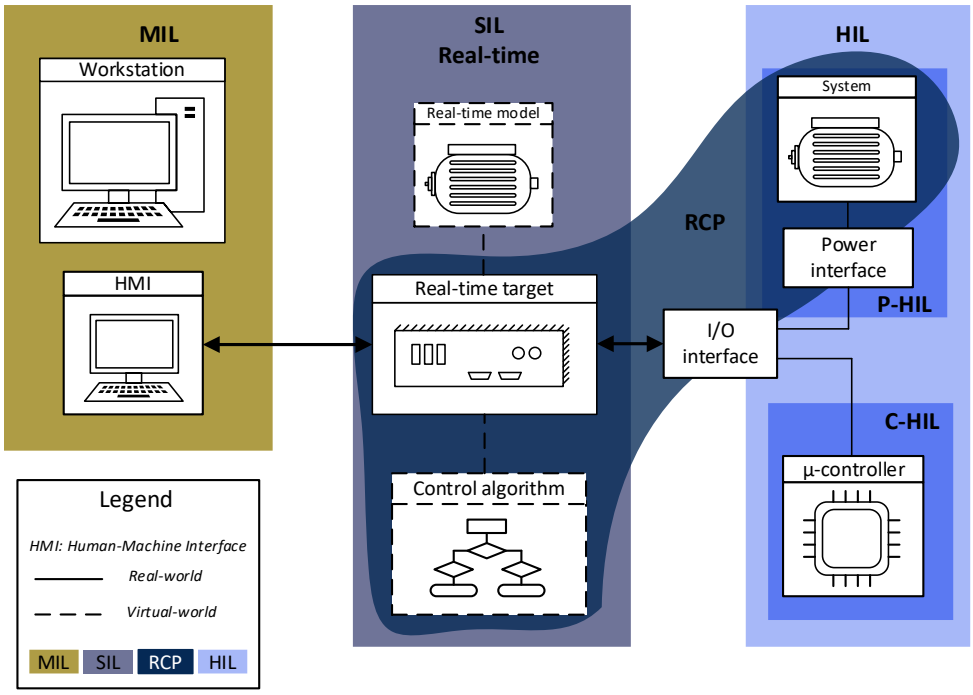

Figure 2. Academic view of the XIL methodology. Dashed lines stand for the virtual world.

2.2. X-in-the-Loop Stages

We will now provide a more detailed explanation of the various components of the general XIL approach in the following sub-sections.

2.2.1. Model-in-the-Loop (MIL)

The MIL approach involves using digital model development to support understanding and functional development work. The interaction between the system, the controller, and the environment run by computation of any solver. System requirements are tested on a virtual environment and any modification of the system specifications can be made at this stage.

This digital phase serves multiple purposes. It allows for system conceptualization through modelling, which is referred to as a model-based design [17,18]. This stage can also be used for performance evaluation [19], sizing, or optimisation [20,21] by testing different configurations or model parameter settings [16,22]. Finally, the digital approach enables the development and assessment of system control laws without any risk of damage or deterioration to the real system [23].

Note that at this stage, the real-time is not a requirement. Simulation can be either faster or slower than real-time. In very accurate physical modelling (including 3-D models), it is common for the computation time to not meet the real-time requirement.

2.2.2. Software-in-the-Loop (SIL)

The SIL phase involves all operations necessary to adapt the model for real-time execution. It is generally overlooked because it does not provide direct benefits. However, it is essential in the overall workflow, as it allows for the proper consideration of hardware constraints when transitioning from MIL to HIL. While MIL is focused on modelling and analysing simulation results, with variable or fixed time steps solvers, SIL encompasses all steps required to adapt models for real-time operation. Indeed, some models can be resource-intensive, especially 3-D finite element models. Therefore, model reduction becomes necessary [24]. This step requires knowledge of both physics to apply sufficient reductions without affecting the reliability of the simulation for the specified objectives and

an understanding of solvers operations to configure them correctly, for instance, by using fixed time steps and adapting the time step relative to the system dynamics. At this stage, we also verify the correct code generation before deploying it on the real-time target [25]. Finally, some also include code deployment and initial testing on a real-time target in this step [26]. It is then important to understand hardware limitations and address them, which involves verifying and analysing CPU load and breaking the model into several parts that can run on multiple cores [27].

There are three ways for a model to meet the real-time requirement as represented in Figure 3. The figure describes three paths for adapting simulation model for meeting the real-time requirement. ROM (Reduced Order Model) strategies can be classified into three main approaches. Data-based methods, including regression models, artificial neural networks (ANNs), and dynamic mode decomposition (DMD), rely on learning system dynamics from historical or simulation data. These are particularly effective for capturing non-linear behaviour when sufficient data is available. Physics-based methods focus on simplifying the governing equations of the system using techniques such as Proper Orthogonal Decomposition (POD), balanced truncation, and state-space linearization, ensuring physically interpretable models. Finally, hybrid approaches, such as Physics-Informed Neural Networks (PINNs) from [28], surrogate models with embedded physics constraints, and Kalman filters augmented with machine learning, combine the strengths of both paradigms to achieve accuracy and efficiency. These ROM techniques enable real-time capabilities by reducing computational demands while retaining essential system dynamics. A complete review of methods analysing the trade-off between models fidelity and computational effort, with uncertainty integration implications in optimisation-oriented applications, is presented in [29].

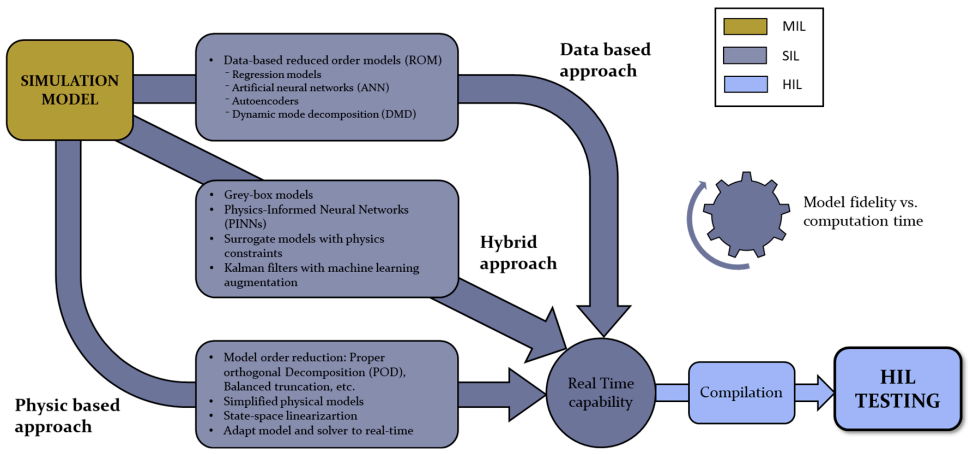

Figure 3. Paths for enabling real-time calculation.

2.2.3. Hardware-in-the-Loop (HIL)

Notable manufacturers of real-time computers include Speedgoat, OPAL-RT, Typhoon HIL, National Instruments, dSPACE. They offer fast computing with operating systems specially tuned for real-time capabilities. They also provide a variety of inputs/outputs configuration interfaces (CAN, Ethernet, Modbus, Serial, USB, Digital/analogue inputs/outputs, etc.).

1. Controller-Hardware-In-the-Loop (C-HIL)

In C-HIL, the controller algorithm is deployed on a real dedicated hardware such as industrial microcontroller or Programmable Logic Controller (PLC). It interacts with a model of the plant running on a real-time target by emulating the physical input/output

interface. We can note that this corresponds to the classical HIL test, from the automotive point of view, where the only hardware tested is the controller.

The main objective of the C-HIL approach is to test algorithms and the physical controller in a controlled and safe environment. This step helps mitigate risks in system control by identifying potential unforeseen modes, for example, due to communication constraints not identified during earlier studies [30]. By controlling the environment, this approach also allows testing beyond what could be achieved in a real system. Indeed, by enabling the emulation of extreme failure scenarios, C-HIL ensures that the algorithms can perform fallback actions [31] or avoid causing more severe damage [32].

2. Power-Hardware-in-the-Loop (P-HIL)

In P-HIL, at least part of the system is tested with real components. These components can be a complete system [33], part of a system [34], actuators [35], and sometimes a physical emulation of the behaviour of an actuator [36] or an energy flow [37].

The P-HIL stage serves as a tool for prototype development by targeting the most critical parts and emulating the interfaces and behaviour of the rest of the system. By extension, unit component testing is also performed at this level [38] and is called Equipment Under Test (EUT), Device Under Test (DUT), System Under Test (SUT), or Hardware Under Test (HUT) [39]. Finally, it enables a final risk-reduction step before integration into its final environment, with the implementation of a demonstrator for a complete system in an emulated environment [33,40].

3. Rapid-Control-Prototyping (RCP)

RCP is close to P-HIL, as it relies on real components [41]. However, it differs significantly in terms of the study focus and the objectives inherent to the approach. Indeed, RCP focuses exclusively on the development of control algorithms.

As its name suggests, this method is centred on control development and offers a fast-track solution for it. This implies that the system to be controlled already exists and that a real-time target, whose operation the user fully understands, is available [42]. It is particularly well-suited for improving existing control systems [43], especially in the field of power electronics [44]. However, it can also pose a risk by acting directly on the real system without a prior validation phase. This is why preliminary tests in simulation, i.e., in MIL, are usually conducted before using RCP [31–34].

2.3. Workflow and Objectives

As part of a complete development process, from concept to finalised prototype, the XIL approach unfolds through the stages MIL → SIL → HIL. The diagram in Figure 4. illustrates this principle, highlighting the sequence and the physical platforms supporting these stages. These platforms are either a computational workstation, dedicated to all types of simulation, a real-time target, running real-time simulation, or real components and environment. The latter are connected to real-time targets through interfaces emulation. Nevertheless, this process is not a strict serial process as it is demonstrated in Figure 4. For example, it is possible, depending on the project specificity, after an MIL step, to go directly to the RCP testing if a prototype is already available and if the objective is to test control laws on the DUT. The figure presents, for each step of the XIL process, the concept and describes the system involved including the controller, the plant, and the environment.

The XIL approach represents a development flow with several stages, each of which includes validation tests that can lead to revisiting the design of the previous stage if inadequacies are identified (for example, a real-time constraint might prevent the developed code from being used on the available target). This approach is aligned with the V-model, which

allows for returning to a previous step. The diagram in Figure 1 attempts to graphically represent the concept of nested V-models.

However, because academic research does not necessarily integrate into a complete process of industrial product design, the XIL approach should be seen as a toolbox that researchers can rely on to achieve various objectives. Thus, a scientific project will utilise one or two parts of the XIL approach depending on its objectives, which can range from exploratory development through MBD (Model-Based Design) to the development of a demonstrator, or to the validation of a proof of concept, and many other objectives. Some specific use cases are listed in Table 2, which summarises the various objectives of each part of the XIL.

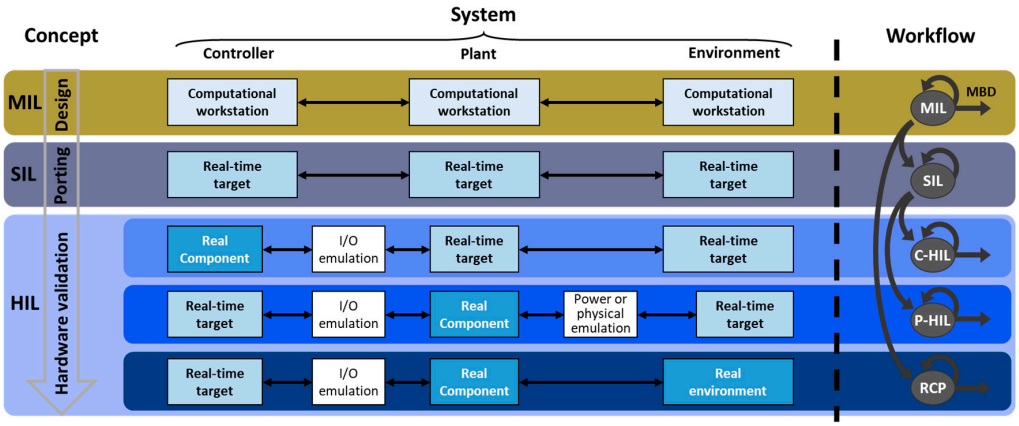

Figure 4. Detailed view of the workflow from the concept design (MIL) to the hardware validation (HIL and RCP). Computational workstations run all kinds of simulation, while real-time targets run only real-time simulation.

Table 2. A non-exhaustive list of specific use cases and the steps used for testing (optional stage in brackets).

Objectives	XIL Academic Approach
Developing a prototype for the control of an existing system	MIL → RCP
Improving/tuning the control of an existing system	[MIL→] RCP
Developing a system from design to implementation	MIL → SIL → C-HIL [→P-HIL]
Integrating and testing an existing algorithm in a controlled environment	C-HIL
Optimising the sizing of an energy chain	MIL
Designing a system and its control using Model-Based Design (MBD)	MIL
Verifying the applicability of an algorithm in real-time	SIL
Performing component tests	P-HIL
Evaluating the performance of a system and its control	MIL [→RCP]
Evaluating the behaviour outside of operating conditions	MIL [→C-HIL]
Verifying the physical behaviour of a system before its final integration	C-HIL + P-HIL

3. Contributions of XIL Methodology to the FCSs Development

This paper is based on a systematic review of the scientific literature from two databases: Scopus and Web of Science. A comprehensive review was conducted on full-text articles published in English exclusively between 2006 and 2024. The search focused on articles containing the keywords “PEMFC” and “Hardware-in-the-Loop,” as well as related

abbreviations such as "HIL", within the title, keywords, and abstract. Based on the current literature analysis, seven identified topics are presented in the following sections.

3.1. Fuel Cell System Modelling

Fuel cells are complex multi-physical systems. They involve simultaneous reactions in multiple physical fields at different scales: electrochemical reactions, thermal production and exchange, and production and movement of water from atomistic scale to system level. These phenomena are intrinsically coupled with different dynamics [25]. FCS models can be very specific and require a long computation time as they can model various phenomena in 2-D or 3-D. However, there are 1-D or 0-D models that require fewer resources for calculation. Depending on the case, multi-physics and dynamic modelling is also required. In [45], at a system level, some of these phenomena are modelled through 1-D models running at different dynamics on separated CPUs and cross-communication between models. For transport applications the high dynamic of the power demand is the main driving factor for PEMFC degradation. Some consequences of this are water management and reactant starvation issues [46,47]. In [48], the authors reviewed durability testing, causes, consequences, and mitigation methods for enhancing fuel cell lifetime. An estimation method of degradation mechanisms using multi-scales and multi-physics modelling approaches is proposed in [49].

It is worth analysing how authors use specific models for improving the fuel cell control and adapt it for real-time integration.

In [50], He et al. study the simplification of fuel cell models for integration into a system controller. Assumptions are presented to simplify the physical equations, and a method for the rapid resolution of the equations is developed. A dynamic model of a PEMFC stack is described, providing an estimation of local parameters inside the stack. Its low computation time makes it suitable for an implementation in HIL test platforms. The linearizing model is also used for real-time modelling of air loop systems as in [51]. The model is fitted for different operating points. It is then used to formulate a control algorithm implemented in an HIL test platform. While the difference between the linearized model and the non-linear model in terms of precision remains moderate, the computation time allows the integration in a controller. Focusing on the computation time of the fuel cell model, Kravos et al. [52] integrate PEM water management in their model dedicated to real-time control. They validate the model with experimental results of a 25 cm^2 single cell, tested at 60 °C for two operating conditions. It is able to predict the liquid water behaviour inside the stack, and its real-time readiness at 1 kHz is demonstrated.

These three examples illustrate how models can be reduced for real-time implementation by focusing on the description of some selected crucial phenomena and used in the fuel cell control. We can note that these are major concerns in fuel cell modelling when modelling specific phenomena, as they have an impact on the fuel cell system itself. Therefore, in order to incorporate them in real-time testing, model reduction is sometimes necessary.

3.2. Power Electronics: FPGA Integration

In transportation applications, power converters typically enable the voltage elevation between the fuel cell and the DC bus. Power converters modelling requires high-speed computation. Compared to CPU (central processing unit), a field programmable gate array (FPGA) offers a better solution for high-speed calculation. Indeed, FPGA can compute power converter modelling at hundreds of nanoseconds rate, whereas CPU offers hundreds of microseconds rate. Hence, fast real-time models of power converters can be tested before prototype development, thereby reducing costs and risks [53]. Several examples of an HIL test platform for power converter models coupled with fuel cell models are given in [54,55].

Additionally, to extend the lifetime of the fuel cell, the general strategy is to reduce the fuel cell current ripple by appropriate control and/or topologies.

A dynamic and transient power smoothing technique is presented in [56] using the virtual inertia concept. In that work, power converter modelling and control are addressed for multi-stack PEMFC systems. Zhou et al. [57] developed a slicing mode control of a DC/DC power converter in order to reduce the current fluctuation of the fuel cell coupled with a supercapacitor. To reduce current ripple and stabilise the output current, Hao et al. [55,58] modelled, prototyped, and controlled a four-phase interleaved boost converter. In [54], Wang et al. developed a six-phase interleaved boost converter that enables fuel cell lifetime extension and a reduction in the weight and volume of magnetic component compared to two other topologies. The HIL test bench validated the topologies and the control in a real-time environment with a real converter prototype.

In these different examples, the XIL methodology for power electronics, is used to simplify and accelerate the design, in order process, allowing researchers to focus on the most relevant subjects of study. These topics are the close control loops of power converter, and the XIL methodology enables the rapid implementation on real hardware.

3.3. Air Supply Management System

The air supply system is responsible for supplying air to the fuel cell with appropriate pressure, mass flow, humidity, and temperature. The air supply management system regulates the air supply system actuators, such as the compressor, in order to improve the fuel cell's efficiency, stability and robustness [59], by providing an optimal oxygen excess ratio (OER) and pressure. Advanced control techniques are developed, first in simulation to validate the benefits of the controller, and then integrated into real-time validation experimentation such as RCP, C-HIL, or HIL test benches, as presented hereafter.

A few works are focused on the control of the air compressor for optimising the OER. Beyond the validation of the real-time application of algorithms, the objectives are the optimisation of the system efficiency and the protection of the fuel cell and the compressor itself. In most of the following examples, real compressors are integrated in test benches. Hence, Matraji et al. [60] developed a RCP test bench around a real compressor and implemented a strategy in order to improve the fuel cell's net power according to the OER. A LabVIEW programme in real time emulates the fuel cell and the control algorithm is running on a CompactRIO controller. In parallel with this work, Laghrouche et al. [61] developed a constrained extremum seeking control algorithm aiming at maintaining the OER and pressure between given limits during dynamic operation of the system. They demonstrated with an HIL test bench integrating a centrifugal compressor that the control could avoid both oxygen starvation in the stack and detrimental operating conditions for the compressor, such as pressure exceeding the surge line. Phan et al. [62] developed a compressor controller validated with a 100 W PEMFC. The controller is designed to maintain an optimal OER under load variations. Wang et al. [63] developed a controller for OER and pressure based on an artificial neural network in order to improve the net output power of the fuel cell system. The RCP test bench is composed of a real compressor with sensors communicating with real-time models of the remaining part of the air supply system running on a dSPACE target. The same setup was used in [64] with a fuzzy logic controller. In [65], a low-level control strategy is designed using a dedicated HIL test-bench with PEMFCs emulated directly using FPGA code. The main technique is non-linear observer-based control of the OER. A complementary approach to low-level control with disturbance rejection was proposed in [66].

Other authors have a more global view of the cathode environment control. For example, in [67], authors developed an HIL test bench for the control of pressure and mass flow at

the cathode. The originality of this works is the use of a virtual state observer for pressure and flow estimation embedded on an Arduino card. Ramos-Paja et al. [68] developed a strategy for minimising the fuel cell output current by acting on the compressor voltage. The compressor voltage controller is compiled on a FPGA controller, with a real-time model of the fuel cell running on a workstation. Guo et al. [69] developed a fault-tolerant controller with an adaptive surface response.

As the compressor is the main power consumer of the fuel cell system, the challenge lies in developing advanced control techniques to maximise its efficiency while ensuring robust and fast control. The emergence of MSFCSs also requires an adapted control of the air loop as several configurations are possible.

3.4. Hydrogen Supply Management System

The hydrogen management system is responsible for the conditioning of H_2 supplied to the stack from the tank. It comprises pressure and flow sensors, and actuators such as valves and regulators. Authors addressing this subsystem use emulated fuel cell stacks and primarily aim at validating the real-time operation of the controller.

Quan et al. [70] developed a controller based on a model predictive control (MPC). The controller manages the H_2 excess ratio (HER) and the anode pressure and reduce overshoot compared to the reference MPC. In order to validate the real-time capability of the proposed controller, a C-HIL test-bench is developed with two DSP boards. A real-time model of the fuel cell system runs on one board, and the controller runs on the other board while data acquisition and visualisation are collected through a CAN bus and transmitted to a desktop computer. Wang et al. [71] developed and validated a controller based on an ANN for the anode pressure and HER control. They used an RCP test bench with real sensors and actuators, a real-time model of the H_2 supply system running on a dSPACE target, and the ANN-based controller running on an ETAS controller. Abbaspour et al. [59] designed a fault-tolerant controller based on an ANN for the fault detection and a non-linear controller for the pressure regulation on the hydrogen and air regulation valves. The controller is validated on a real-time model of the PEMFC system and capable of maintaining system stability in case of air exhaust and H_2 regulation valve fault. They are both running on a real-time OPAL-RT target.

On this topic, we notice that several authors communicate on advanced techniques using artificial intelligence, but more simple control techniques are not reported. Although simple control techniques, including PID controllers, are generally effective, it is noteworthy that there is a surprising lack of publications on the evaluation of classical controllers in a Hardware-in-the-Loop (HIL) environment.

3.5. Thermal Management System

The thermal management system is responsible for maintaining the fuel cell stack temperature within an acceptable range, and typically no more than 80 °C. Typical architectures of this subsystem include, pumps, radiators, fans, and sensors. The particularity of the thermal system is its low dynamic compared to the electrical and gas systems.

In order to integrate a reliable fuel cell model into the energy management controller, He et al. [72] developed a predictive model that includes stack temperature estimation. The application of this approach for thermal management is validated with HIL experiments, emulating both automotive and stationary applications. Compared to rule-based strategies, the EMS including this model allows up to 10 % of performance improvement. The thermal management system can also include a fault-tolerant control, as developed in [73] for a multi-stack fuel cell system. With this approach, HIL results show that in case of pump

fault injection, the reference temperature is reached again after only 20 s. However, there is no information available concerning the experimental setup in this reference.

As shown in the previous section, the literature lacks studies on simple control techniques. However, as the nominal power of MSFCSs continues to increase, more complex and demanding thermal management strategies will be required. In this context, multivariable and optimal control approaches could play a crucial role in ensuring optimal performance under varying operating conditions.

3.6. Energy Management Systems

Energy management is responsible for power balancing between energy sources in order to respond to the demand in a safe and efficient way. It takes into consideration several phenomena: hybridization with batteries and/or supercapacitors, fuel cell degradation, fuel cell efficiency, hydrogen consumption, DC bus voltage regulation, etc. In a real-world application, it combines sensors, actuators, algorithms, programmable devices, and display devices into an energy management system (EMS). After validating the interest and capability of the EMS on MIL step, the next step is to compile and export it to a dedicated controller for HIL tests (including C-HIL and P-HIL test) for the validation on a real hardware environment. We can distinguish between offline and online EMS optimisation. In the first case, the mission profile is known in advance and real-time capability is not required. The objective of offline optimisation is to compute optimal energy management in order to have the best path according to the objective. In contrast, online control requires real-time capability for the control of the system. In this case, the mission profile is not known or partially known in advance. Consequently, real-time capability and robustness are validated in this step, using HIL and RCP testing control. Rule-based or frequency-based [74] controllers are easy to design, robust, and have a fast computation time. However, modern control techniques, including model predictive control (MPC) [75] or equivalent consumption minimisation strategy (ECMS) [76], allow further enhancements of the EMS.

A first class of EMS studied by several authors up to HIL testing is rule-based, integrating some specific enhancements. For instance, in [77] a rule-based EMS is proposed using low-pass filtering. This is combined with ageing modelling of hybrid PEMFCs and battery system to properly size the hybrid system components. Evaluation on an HIL test bench is proposed with the real-time target integrating the vehicle physical model and the controller implemented on a dedicated vehicle control unit. Similarly, Nazir et al. [78] developed a two-level controller for the power splitting and close control of a hybrid fuel cell and supercapacitor system. The power splitting between the fuel cell and the super-capacitor is made by a rule-based control. The close control of the fuel cell and super-capacitor currents use a backstepping non-linear controller that is adaptive with hardware parameters time variations (inductors, capacitors, and resistors). They compiled the proposed controller on a microcontroller board that communicate in real time with the fuel cell and super-capacitor models (Simulink models) on a desktop computer. Experimental results are in accordance with the MIL results in order to stabilise the DC bus. In [79] authors enhance the online control made by a lookup-table for the power split between fuel cell system and a battery by an offline optimisation using mixed integer linear programming (MILP) and a Bayesian optimisation. The optimisation allows the reduction of the H_2 equivalent consumption by 25 % and an HIL platform was built in order to validate the real-time capability of models and control.

In another class of EMS, controllers integrating an algorithm for the minimisation of the hydrogen consumption of the system are proposed. Hence, Li et al. [80] developed a controller based on a combination with a state machine and an equivalent consumption

minimisation strategy (ECMS) for a hybrid tramway combining a fuel cell with a supercapacitor. The proposed control is tested on an HIL test bench. The system model runs on an OPAL-RT target and the proposed control is running on a DSP board. In [81], an ECMS approach is proposed for an optimal online EMS on a hybrid powertrain architecture. The optimal approach is compared with a more basic rule-based technique and it is evaluated on HIL test bench with the controller implemented in a dSPACE MicroAutoBox device. Yan et al. [82] developed a strategy for minimisation of the equivalent energy consumption based on the state of the tramway (traction, braking, coasting, or parking). Their control algorithm is tested in real time on OPAL-RT target. The system model runs in real time on a CPU of the target and the control on another CPU of the same target.

Other studies propose integrating some learning capability in the EMS algorithm. In [83], an online EMS based on reinforcement learning, specifically Q-learning, is designed and validated on a dedicated HIL test bench. In this type of approach, the HIL testing allows the provision of insight into the online feasibility of machine learning algorithms containing inference methods. The inference steps should be fast enough to properly provide efficient power flow management for automotive powertrain applications. In [84], Kandidayeni et al. studied an EMS dedicated to improving the efficiency of the hybrid powertrain. Their controller performs an online recognition of driving patterns. The system efficiency is enhanced compared to a fuzzy-logic controller integrating offline pattern recognition, obtained through offline optimisation by a genetic algorithm. The largest improvement is achieved when the fuel cell undergoes a performance drift. Shi et al. [85] developed an EMS based on machine learning in order to reduce online the hydrogen consumption of a hydrogen fuel cell vehicle (Toyota Mirai). They test and validate the ability to reconfigure in case of stack loss by a C-HIL test.

Some authors also propose predictive control for the EMS and validate their approach through HIL testing. Pereira et al. [86] developed a non-linear predictive controller in order to find the maximum efficiency current point of the fuel cell and minimise its degradation. They validated the controller on a P-HIL test bench including a 3 kW PEMFC. The EMS runs on a CPU in a small-sized computer that also controls the electronic load. An adaptive online optimisation based on an extremum seeking is developed by Zhou in [87]. The fuel cell operates in the maximum efficiency area and the SOC of the hybridization battery stays within the predetermined interval. The proposed controller is first tested on a C-HIL test bench combining the SCALEXIO and MicroLabBox modules from dSPACE manufacturer (Paderborn, Germany). Secondly, a 1.2 kW PEMFC system is introduced for RCP testing.

Future EMS development should account for time-varying parameters, such as the degradation of fuel cell's state of health. However, the different degradation mechanisms remain an active area of research, and their integration with modelling advancements requires further investigation. This includes more accurately calibrated physical degradation models to enhance predictive capabilities.

3.7. Multi-Stack FCS

A multi-stack fuel cell system (MSFCS) is an association of several (two or more) fuel cell stack in series or in parallel as presented in Figure 5. This figure presents four possible electrical architecture for multi-stack fuel cell associations with power converters. The association of several stack in an MSFCS allows us to reach high power levels, typically above hundreds of kilowatts, that are not achievable with a single stack. Therefore, power splitting between stacks must be performed. Using several stacks as independent modules of an MSFCS allows the resilience of the system in case of failure, as well as the possibility of selective start-up and shut-down in order to reduce consumption and degradation [88,89].

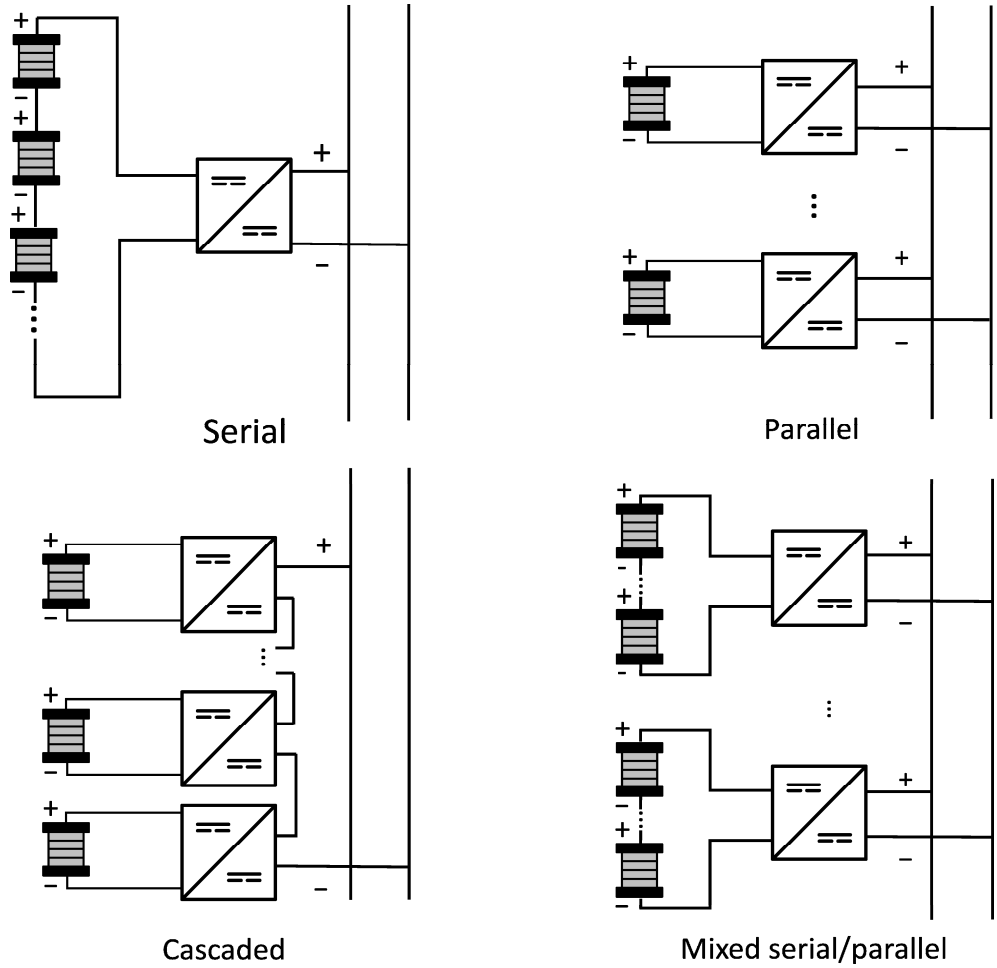

Figure 5. Possible electrical configurations of MSFCSs with power converters.

The main issues encountered in the literature are related to architecture, power conditioning, and EMS. HIL testing is particularly appropriate for the study of MSFCS control due to the high cost of these systems. Detailed reviews of the subject have been conducted by [90,91]. As a summary, we describe the most relevant EMS studies selected in the literature. It can be noted that the enhancement of MSFCS lifetime is a major concern in MSFCS EMS [92]. Another main concern is the development of EMS with fault-tolerant and reconfiguration properties.

A key challenge in MSFCSs is the design of architectures, which is not actually in the final stage of XIL methodology, as it is a combinatorial problem. There is a variety of possibilities from electrical to fluidic structures [91]. A complete fluidic architecture is presented in Figure 6 and is extracted from a technical report of the American Department of Energy [93]. This figure depicts the air, hydrogen, and coolant architecture with actuators and sensors for a heavy-duty vehicle containing two fuel cell stacks. As with control laws design, XIL testing provides a safe way to test architectures. Different architecture models can be proven on HIL testing to select the best architecture and develop an appropriate control.

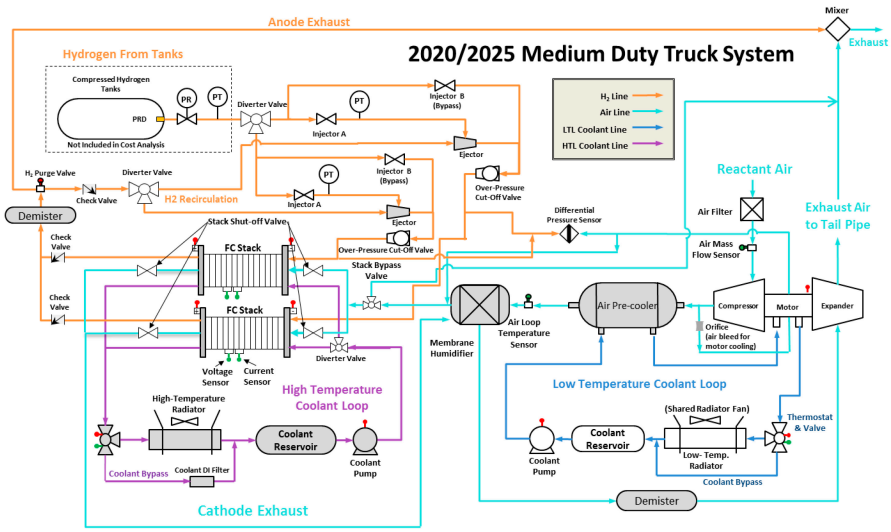

Figure 6. Fluidic architecture for multi-stack heavy duty transportation (extracted from [93]).

Several studies aim at extending the lifetime of the stacks. To address the power split in an MSFCS, Meng et al. [94] developed a consensual EMS. The control is able to increase the lifetime of the system by 36 % through a consensual repartition of the load between the stacks in order to equilibrate fuel cell degradation and then reach the end-of-life of each individual stack at a similar time. The EMS is also capable of supplying the load even in the case of electrical or communication failure. They tested in real time on an HIL test bench the capability of the controller. The controller runs on a DSP board while the MSFCS real-time model runs on an OPAL-RT target. Wang et al. [95] developed a power allocation method based on virtual droop control for an MSFCS integrated into a tramway application with the ability to uniform the degradation among the stacks in order to extend the lifetime of the MSFCS. The power allocation method is also capable to redistribute the power load between the remaining stacks in case of one stack failure. To extend the lifetime of the fuel cell stacks, they chose to reduce to power fluctuation of the degraded stacks. To validate the EMS, they build an HIL test-bench with real-time models running on an OPAL-RT target and the controller running on a Texas Instruments controller. Results show that the proposed EMS is able to follow the power demand, even in the case of one stack failure, and to reduce the power fluctuation of the more degraded stacks, leading to a standardisation of the degradation and to an extend of the lifetime of the whole system. In [56], the EMS approach is based on the concept of virtual inertia and droop-based control to mitigate power transients among the different stacks operating in parallel. The approach is also adaptive by varying the virtual inertia according to each FC stack's degradation trend. This results in a power-smoothing algorithm with performance evaluation on an RT-LAB HIL test platform.

Real-time simulation of the MSFCS is used by several authors for a detailed analysis of the behaviour of the controller. Peng et al. [96] proposed a simulation called HIL with the controller not exported to an embedded target, but rather running on a real-time computer (dSPACE). There are only real-time models running on the real-time computer. Therefore, according to the proposed terminology, they follow a SIL approach. Nevertheless, they develop a two-stage EMS strategy in order to reduce fuel consumption and study the impact of the coordination between top-level and low-level management systems. Jian et al. [97] presented an HIL real-time validation of the rule-based control algorithm dedicated to a fuel cell vehicle equipped with four stacks hybridised by a battery. Thanks to the real-time emulation of the system, they were able to verify the stability of the bus voltage during a standardised driving cycle. Their approach could be employed to test the redundancy

of the system, which is theoretically improved thanks to the number of stacks. However, some improvements to the models and of the control algorithm are needed, in order to integrate fuel cell degradation, and prognostic and health management algorithms.

This topic brings several difficulties that the XIL methodologies could help to address. From MIL to HIL testing, the XIL methodology is particularly suited to enhance EMS, architecture, and degradation issues at a reduced cost.

3.8. Synthesis of the Literature in the Field

A growing number of articles have been published in the field of PEM fuel cells, and among them, publications focusing on HIL methodology are becoming increasingly numerous. In order to make a statistical study of the literature in the field, extractions from the Scopus and Web of Science databases have been made. The following requests were applied: first, ("PEMFC" OR ("Fuel cell" AND ("MEMBRANE" OR "PEM"))) in title, abstract and keywords, and secondly completed with AND ("Hardware in the loop" OR "Hardware-in-the-loop" OR "HIL"), depending on whether the search concerned PEM fuel cells globally or regarding specifically HIL methodology. The requests are limited to full articles written in English between 2006 and 2024 (inclusive). The evolution of the number of publications during this period is plotted in Figure 7a. This figure shows the number of publications focusing on HIL in blue bars and the fraction of the HIL publication in the PEMFC field in a red curve. This demonstrates that, after more than a decade with a low and variable number of publications mentioning HIL methodology for PEMFCs, a progressive growth started in 2018, with a notable acceleration in 2024. Concurrently, the fraction of these publications among the total number of publications dealing with PEM fuel cells also shows a significant growth in 2018, with an acceleration in 2024, from 0.03 % in 2017 to 0.71 % in 2024. Hence, even though this particular topic still remains a minor one for the fuel cell community, its importance has grown significantly in the last few years.

Table 3. Synthesis of the literature of XIL testing for PEMFCs.

Topics	Stage	Remarks	System Under Test	Reference
Fuel cell system modelling for real-time capability	SIL	Model reduction in complete FCS.	None	He, 2023 [50]
	SIL	Development of a linear model for air loop control.	None	Hu, 2023 [51]
	MIL/SIL	Water management modelling. 1 kHz capability.	None	Kravos, 2022 [52]
Power electronics	HIL	Coupling FPGA for power electronics modelling and CPU for fuel cell modelling.	None	Ma, 2018 [53]
	HIL	Converter modelled in FPGA board of a MicroLabBox, controller, and fuel cell model running on the real-time processor of the MicroLabBox.	Power converter	Wang, 2024 [54]
	C-HIL	Power converter control development for an MSFCS. Control is running on a DSP.	DSP controller	Li, 2023 [56]
	HIL	Power converter control. Two dSPACE are used for modelling and control.	None	Zhou, 2021 [57]
	P-HIL	Power converter control and integration of a 1 kW fuel cell. Controller deported on dSPACE and a FPGA board.	FC + Controller	Hao, 2021 [55] and [58]
	HIL	Propose and test different methods for computation time reduction in an RT-LAB HIL test-bench.	None	Jung, 2010 [98–100]

Table 3. *Cont.*

Topics	Stage	Remarks	System Under Test	Reference
Gases supply	HIL	Fault tolerant control for air and H_2. Everything is tested on an OPAL-RT target.	None	Abbaspour, 2019 [59]
Air supply	RCP	Control of the OER for improving the net power efficiency.	Compressor	Matraji, 2013 [60]
	RCP	Control of the OER and pressure under dynamic load.	Compressor	Laghrouche, 2013 [61]
	HIL	State observer for oxygen flow and pressure running on an ARDUINO communicating with real-time fuel cell model.	ARDUINO state observer	Olteanu, 2015 [67]
	RCP	Control of OER under dynamic load. Use a 100 W fuel cell for validation of the control algorithm.	FC + compressor	Phan, 2023 [62]
	C-HIL	Control of the compressor voltage running on an FPGA. Real-time model running on a computer.	FPGA controller	Ramos-Paja, 2014 [68]
	RCP	Control of OER based on an artificial neural network. Models and controller run on dSPACE. A real compressor is tested.	Compressor	Wang, 2023 [63]
	RCP	Control of OER based on fuzzy logic. Models and controller run on dSPACE. A real compressor is tested.	Compressor	Zhang, 2020 [64]
	C-HIL	Adaptive fault tolerant controller for the OER. The model is running on a Speedgoat while the controller is compiled on a DSP.	DSP controller	Guo, 2024 [69]
	RCP	Adaptive control of the OER based on a state observer. FC model is running on an FPGA. A real compressor is included. A CompactRIO controls in real-time the compressor.	Compressor	Liu, 2014 [65]
	C-HIL	Control of the OER based on a state observer. FC model is running on OPAL-RT. The controller runs on a DSP.	DSP controller	Ma, 2023 [66]
H_2 supply	C-HIL	Model predictive control of the H_2 excess ratio. Two DSP used for modelling and control.	DSP controller	Quan, 2021 [70]
	RCP	Controller based on artificial neural networks for the anode pressure and HER control. Test-bench include real sensors, a motor, and a pump.	Motor, pump and sensors	Wang, 2023 [71]
Thermal management	MIL	3-D finite element modelling of the thermal component on a fuel cell system. MIL used for designing the thermal system.	None	Mayyas, 2014 [101]
	C-HIL	Model predictive controller including stack temperature estimation developed and included in HIL testing.	Controller	He, 2020 [72]

Table 3. *Cont.*

Topics	Stage	Remarks	System Under Test	Reference
EMS	C-HIL	Rule-based EMS using low pass filter and ageing modelling. Controller running on a vehicle control unit communicating with a real-time vehicle model.	Controller	Lu, 2023 [77]
	HIL	Lookup table-based EMS optimised off-line by a MILP and Bayesian optimisation. The control in embedded on a dSPACE target and real-time models are running on a Speedgoat target.	Controller	Lambert, 2025 [79]
	C-HIL	Controller based on state machine and ECMS for a tramway application. Models are running on an OPAL-RT target while the controller runs on a DSP.	Controller	Li, 2019 [80]
	HIL	Minimisation of the equivalent energy consumption based on the tramway state. Model and controller are running on two separate CPU of the OPAL-RT target.	None	Yan, 2019 [82]
	RCP	Online H_2 consumption optimisation by an ECMS technique. Validation of the proposed algorithm on an RCP test-bench with a 1.2 kW fuel cell and the controller running on a dSPACE MicroAutoBox.	Controller + FC	Li, 2018 [81]
	P-HIL/C-HIL	EMS based on reinforcement leaning with online prediction tested on an HIL setup. Real components controlled by a vehicle control unit.	Power train + controller	Lin, 2021 [83]
	RCP	EMS based on a fuzzy logic controller and an online driving cycle recognition. A LabVIEW model considering electric motor, battery, and converters and the EMS gives online current set point for the real 500 W fuel cell system.	FC	Kandidayeni, 2019 [84]
	C-HIL	EMS strategy for hybrid fuel cell and super-capacitor vehicle. Rule-based power splitting and a close control for power converter with the integration of the time-varying parameters such as inductance and capacitance. The controller is tested on a C-HIL setup with models running on a desktop computer.	Controller	Nazir, 2020 [78]
	C-HIL	Non-linear predictive control of the fuel cell current in order to find the maximum efficiency point. Test-bench include a 3 kW fuel cell. Controller is running on a BeagleBone Black processor.	Controller	Pereira, 2021 [86]
	C-HIL/RCP	Online optimisation based on fractional order extremum seeking. Online results compared with offline optimisation by dynamic programming. RCP include a 1.2 kW fuel cell.	Controller + FC	Zhou, 2018 [87]

Table 3. *Cont.*

Topics	Stage	Remarks	System Under Test	Reference
EMS for MSFCS	C-HIL	Consensual power splitting of MSFCS for extending the system lifetime. Load following even in the case of electrical failure. Controller validated on a DSP communicating with models on an OPAL-RT target	Controller	Meng, 2021 [94]
	C-HIL	Virtual droop control of the MSFCS in a tramway application. Power splitting and load following even in case of electrical failure. Validation on a C-HIL test-bench.	Controller	Wang, 2020 [95]
	C-HIL	Adaptive virtual droop control depending on each stack degradation. The controller is running on a DSP while models are running on an OPAL-RT target.	Controller	Li, 2023 [56]
	C-HIL	EMS based on machine learning in order to reduce online the hydrogen consumption of a commercial hydrogen fuel cell vehicle model. They tested and validated the ability of reconfiguration in case of loss of stack.	Controller	Shi, 2022 [85]
	SIL	Impact of the coordination between top-level and low-level management systems on fuel consumption.	None	Peng, 2023 [96]
	C-HIL	Validation of the EMS with a 4 stacks MSFCS with bus voltage stability as goal. Validation of the controller in a CHIL setup.	Controller	Jian et al. [97]

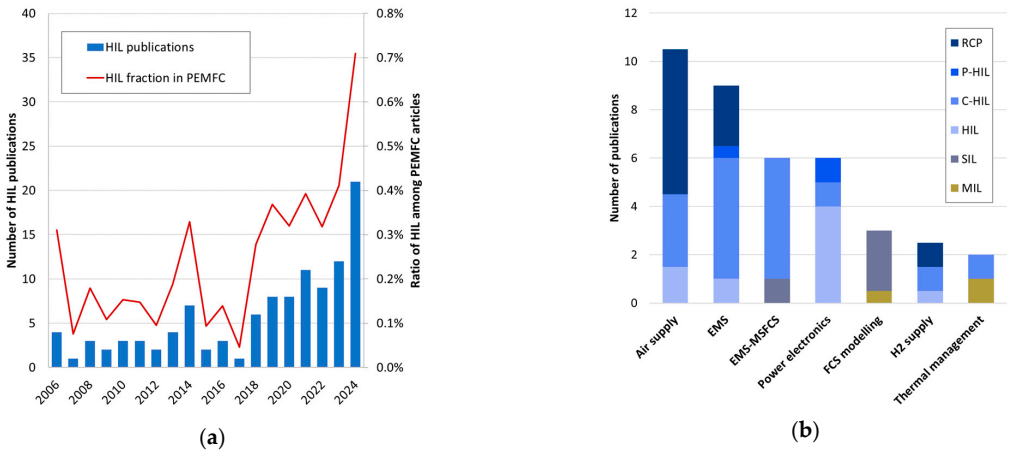

Figure 7. Evolution with time of the number of publications related to HIL methodology for PEMFCs, and of the ratio with the total number of publications about PEMFCs (**a**); classification of the most relevant publications listed in Table 3 (**b**).

For a quick search based on the topic and testing stage, we synthesise the publications selected in the literature in Table 3. It references the topics, the development stage, and some remarks are added for fast comprehension. Its content is summarised in Figure 7b, which shows that C-HIL and RCP are, by far, the most used HIL methods for PEMFC system optimisation. It also confirms that HIL techniques are mainly applied to air supply control and EMS for single-stack and multi-stack systems. However, we found that thermal

management and hydrogen loop are two areas that are not well studied in the literature. Nevertheless, these subjects are also important and should be considered in future studies.

4. Future Challenges

4.1. Modelling

FC modelling often relies on highly detailed electrochemical models to simulate their internal processes. However, fully modelling and understanding of degradation phenomena remain a current challenge and a major research focus. Improving degradation models is essential for predicting fuel cell lifetime and for creating adaptive management systems. XIL methods can be crucial in validating high-fidelity models with real-time data to dynamically adjust parameters, control laws limiting ageing, and refine degradation predictions over time.

To make real-time energy management, control, and optimisation feasible, faster models are required. Such models need to retain key physical insights while reducing computational demand, enabling faster predictions. Developing reduced-order models (ROMs) that incorporate complex electrochemical behaviour is challenging. XIL can facilitate iterative testing and validation of these ROMs, verifying that model simplifications do not compromise critical physical information needed to properly manage a system.

Online-capable models can provide insights into real-time system response and potential degradation impacts, offering advantages for predictive maintenance and adaptive energy management. Machine learning (ML) approaches, particularly physics-informed neural networks (PINNs) and Neural Operators (NOs), can integrate physical laws with real data measurements to capture unmodelled physics, such as degradation phenomena. XIL can help validate these ML models in near real-time environments, using HIL to evaluate emulated responses to real-world conditions and ensure robustness in various operating scenarios.

Significant challenges remain open topics in fuel cell modelling, including limited adaptability to real-time conditions, difficulty in scaling models across varying operating environments, and the high computational demands of detailed models. These factors complicate the real-time application of high-fidelity models. To address these issues, adaptive modelling techniques, such as real-time parameter adjustment and machine learning models, could increase the model's flexibility to handle different scenarios. Hybrid modelling, combining high-fidelity physics-based models with ROMs, could balance accuracy and efficiency, making real-time application more feasible.

4.2. Architecture Optimisation

MSFCSs require complex architectures to handle electrical, thermal, and fluid management needs. With components and solutions such as hydrogen or air tank buffers and common air supplies [102]; for example, there is significant room for optimisation in terms of cost, efficiency, and reliability. Complete methodologies are needed to optimise MSFCS architectures across multiple criteria, including cost, material efficiency, and degradation.

Life cycle analysis (LCA) and emission optimisation are becoming critical in today's climate change context for any system design, where material use and environmental impact play increasingly important roles. Integrating these considerations into architecture optimisation routines ensures that the FCS are designed with sustainability in mind.

XIL enables a flexible environment that allows for iterative test of different configurations of multi-stack architectures by simulating multiple, integrated domains (electrical, thermal, and fluidic) and observing interdependencies under various operating conditions. It provides a testing framework where LCA metrics and emission factors can be directly

integrated into optimisation algorithms, allowing for simultaneous consideration of cost, efficiency, and environmental impact.

The remaining challenges relate to interdependence of component lifespans with system design, mutual subsystem usage optimisation (air/O_2 and H_2 supply), and complex thermal management needs. Proposed solutions include integrating predictive degradation models to optimise component lifespans within the overall architecture, implementing modular designs for hydrogen and fluid systems to increase flexibility, and using real-time digital twins to create more efficient thermal management strategies.

4.3. Energy, Thermal, Power, and Fluid Management Optimisation

Traditionally, electrical energy, thermal management, and fluid management are handled independently. However, FCS could benefit from an integrated management approach, where dependencies between energy, thermal, and fluidic domains are considered and thus enhancing overall performances. XIL allows these interdependencies to be explored through simulations that integrate all management systems, providing insights into how simultaneous adjustments might yield efficiency gains.

Better energy management practices can reduce hardware demands during system design, creating a close interdependence between management strategies and architecture optimisation. XIL allows for concurrent testing of these relationships, helping to identify strategies that balance effective energy management with minimal hardware requirements.

XIL provides an ideal platform for testing classic control algorithms and even advanced techniques, such as learning-based approaches (reinforcement learning and transfer learning) in a controlled environment. SIL, MIL, and HIL configurations can simulate a range of scenarios, including safety issue scenarios, offering a robust environment to validate novel management approaches before implementation on a real system. XIL is crucial for testing and refining communication protocols and the details of energy management system (EMS) algorithms, such as start/stop sequences, supervision logic, and fault detection. By simulating these components, XIL enables thorough validation of communication reliability, data synchronisation, and responsiveness between subsystems, ensuring robust and efficient EMS operations.

4.4. High-Power MSFCSs

The increasing power of fuel cell applications, accompanied by a rise in multi-stack systems (MSFCSs), is bringing new challenges to the fore that are not yet sufficiently addressed in the literature.

Higher nominal power rates raise short-circuit current challenges, particularly as fuel cells lack the inertia found in other power sources with mechanical rotating masses (e.g., electrical generators and motors). An XIL platform may help to better understand the current characteristics of a fuel cell under short-circuit conditions. Current peaks as high as 10 times the nominal values and stable short-circuit currents limited mainly by the mass transfer losses and hydrogen flow were reported in [103] through an HIL test bench. This understanding is essential to ensure reliable electrical protection, selectivity, and essential improve the design of the electrical power system.

The combination of several systems in parallel leads to the study of new electrical architectures, such as some of those presented in Section 3.7. These electrical architectures are associated with power converter topologies, all of which aim to finely control the current of each fuel cell system. Some topologies, such as interleaved boost converters, are well studied in the literature. Their implementation in an HIL-type workflow, where the fuel cell is emulated, is commonly used [54]. Other very interesting topologies, based on power partial converters, are still on the fringe and merit further investigation. A number

of studies have presented this type of topology, which offers a gain in dimensioning [104], but an XIL-type approach is not associated with it. Such an approach would enable us to quickly advance in evaluating and validating these topologies.

If the integration of electrochemical impedance spectroscopy functionalities in PEMFC power converters are now well known [105,106], these studies are rarely carried out with an XIL workflow, which would allow the validation of detection algorithms and control in real-time configurations. The improvement of fuel cell would also be easier with this kind of approach.

5. Conclusions

Fuel cell system development and optimisation are generally supported by various modelling activities. Procedures such as Model-in-the-loop (MIL) and Software-in-the-loop (SIL) help to design the architecture and control strategy of the system using virtual components. On the other hand, Hardware-in-the-loop (HIL) allows for focusing the studies on specific subsystems, combining virtual and real components in dedicated experimental test facilities. These procedures, and some variants, are brought together under the X-in-the-loop (XIL) terminology. The XIL approach, which can be integrated in an industrial V-cycle development methodology, is dedicated to system development and analysis. As its main contribution, this paper presents in a very descriptive way the methodology and how this methodology—or at least parts of it—have been used by academics in order to enhance fuel cells and fuel cell systems. This paper highlights the XIL approach as a key strategy for developing PEMFC system solutions, with a particular focus on its capacity to minimise risks in the complex design and testing phases.

A detailed classification of the procedures included in this methodology is presented in this paper. It offers a complete overview of its use for fuel cell system development, and a clear definition is proposed for each term designating a single procedure to avoid misunderstandings. We observed that the academic sector does not use the entire V-cycle but only part of it.

In this work, we conducted a systematic review of the state of the art. This review allows us to highlight the main topics addressed by the XIL methodology. It appears that air loop management, power converter development for fuel cell application, and validation of EMS in HIL test-bench have a rich literature. In the reviewed studies, the XIL methodology helped to enhance these topics, improving lifetime, efficiency, and robustness in realistic conditions. Conversely, the analysis of the literature demonstrates a lack of research activity on some aspects of balance of plant management, specifically thermal management and hydrogen conditioning.

Furthermore, we identified some modelling topics as major concerns. Particularly, progress is required for the modelling of specific phenomena, such as fuel cell degradation. On this topic, the models need to be adapted to acquire a real-time capability in order to be implemented into real-time control of the fuel cell. In this context, the development of AI techniques for model reduction will help to adapt these complex models for real-time operation, allowing their integration in HIL testing, including virtual fuel cells.

Such progress in modelling seems crucial regarding the optimisation of multi-stack fuel cell systems (MSFCSs). Indeed, the test of these complex system architectures with real components generally leads to high costs, due to FCS costs and the high hydrogen consumption associated. Moreover, the coupling of multiple systems leads to a variety of possible electrical architectures that should be tested individually. These shortcomings can be overcome by the use of the XIL approach and the use of some virtual components, which is increasingly implemented in MSFCS development works.

More generally, the use of the XIL methodology for FCS optimisation should continue to grow, as it appears to be an effective way of capturing the complex interactions between the subsystems in order to achieve an optimal control for a reliable, durable, and efficient system.

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Abbreviations

ANN	Artificial Neural Network
CIL	Controller-in-the-loop
C-HIL	Controller Hardware-in-the-loop
CPU	Central Processing Unit
DMD	Dynamic Mode Decomposition
DSP	Digital Signal Processor
DUT	Device Under Test
ECMS	Equivalent Consumption Minimisation Strategy
EMS	Energy Management System
EUT	Equipment Under Test
FCS	Fuel cell System
FIL	FPGA-in-the-loop
FPGA	Field Programmable Gate Array
HER	Hydrogen Excess Ratio
HIL	Hardware-in-the-loop
HMI	Human–Machine Interface
HUT	Hardware Under Test
LCA	Life Cycle Analysis
MBD	Model-based Design
MIL	Model-in-the-loop
MILP	Mixed Integer Linear Programming
ML	Machine Learning
MPC	Model Predictive Control
MSFCSs	Multi-stack fuel cell systems
OER	Oxygen Excess Ratio
PEMFC	Proton Exchange Membrane Fuel Cell
P-HIL	Power Hardware-in-the-loop
PID	Proportional Integral Derivative
POD	Proper Orthogonal Decomposition
PIL	Processor-in-the-loop
PINNs	Physics-Informed Neural Networks
PLC	Programmable Logic Controller
RCP	Rapid Control Prototyping
RT	Real-time
RIL	Real-physics-in-the-loop
ROM	Reduced Order Model
SIL	Software-in-the-loop
SOC	State Of Charge
SUT	System Under Test
XIL	X-in-the-loop

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